

PLLM: Private Inference and Reproducible LLM Experimentation

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Why PLLM

Ordinary hosted inference protects requests in transit but gives the endpoint access to plaintext prompts and model state. Private-inference research offers alternatives; its results often use different models, hardware, trust assumptions, and metrics. Those results cannot be ranked by latency alone.

PLLM combines a working, **public-weight private-inference path** with an experimentation system for testing compatible research methods on comparable workloads. An ordinary provider endpoint does not become private by changing its URL: the computation and the trust boundary must change.

A private-inference path

The current public-weight path separates three roles:

- **Client** keeps plaintext prompts, tokenization, non-linear and attention state, sampling, and output decoding. A local Responses and Chat Completions gateway can serve existing applications inside this trusted boundary.
- **Preparation** receives one-use seeds before generation, computes masked corrections for the public model's matrix stages, and uploads them to Inference.
- **Inference** holds the public transformer body and evaluates masked integer activations online. It never receives the corresponding plaintext activation.

Client reserves each prepared row once; unused rows burn on cancellation or failure. Preparation is idle during online generation. This is an explicit trust arrangement: Preparation must follow the protocol, erase masks, and **not collude** with Inference. A single operator controlling both services could reconstruct activations. Running all roles on one machine tests functionality, not independent deployment.

A way to test research

PLLM treats papers as provenance, not executable dependencies. Researchers can reimplement a method behind a typed component contract, combine compatible choices into an immutable experiment, search a bounded set of

valid candidates, and benchmark them. Static provider manifests allow discovery without loading provider code; extension is subject to the same capability and execution checks.

Evidence is useful only within a matched cohort: model and checkpoint, workload, warm state, environment, numeric policy, trust assumptions, and metric meaning must be comparable. Search proposes candidates; it does not certify their privacy, output quality, or security. PLLM has **not** demonstrated a state-of-the-art cross-system comparison.

Status and next step

The prepared runtime, local gateway, one-use inventory, and loopback benchmark driver work today. A pinned Qwen2.5-0.5B checkpoint has historical three-role runtime measurements and a separate clear compiled-kernel functionality test. An optional trusted-client Freivalds check has matched **tiny-model** functional evidence, not a real-checkpoint performance study.

Semantic adapters cover additional model configurations, and experimental garbling and cache transforms exist as components. They are **not** a complete protected decoder or proof that those checkpoints run privately end to end. The nine retained Qwen2.5 measurements come from one Apple M5 CPU loopback host; they establish neither WAN/GPU performance nor production non-collusion.

Next: complete protected whole-model execution, measure matched alternatives at realistic scale, and review security and quality claims against that evidence. The technical paper records the protocol, exact cohorts, implementation status, and limits.